

The Room as the Corpus

Four experiments to run with your hands,
before we build the library

Today the corpus is not on a server. It is this room, these tables, and the paper on the floor. Everything we build this afternoon, you will have already done by lunch.

HOW THIS WORKS

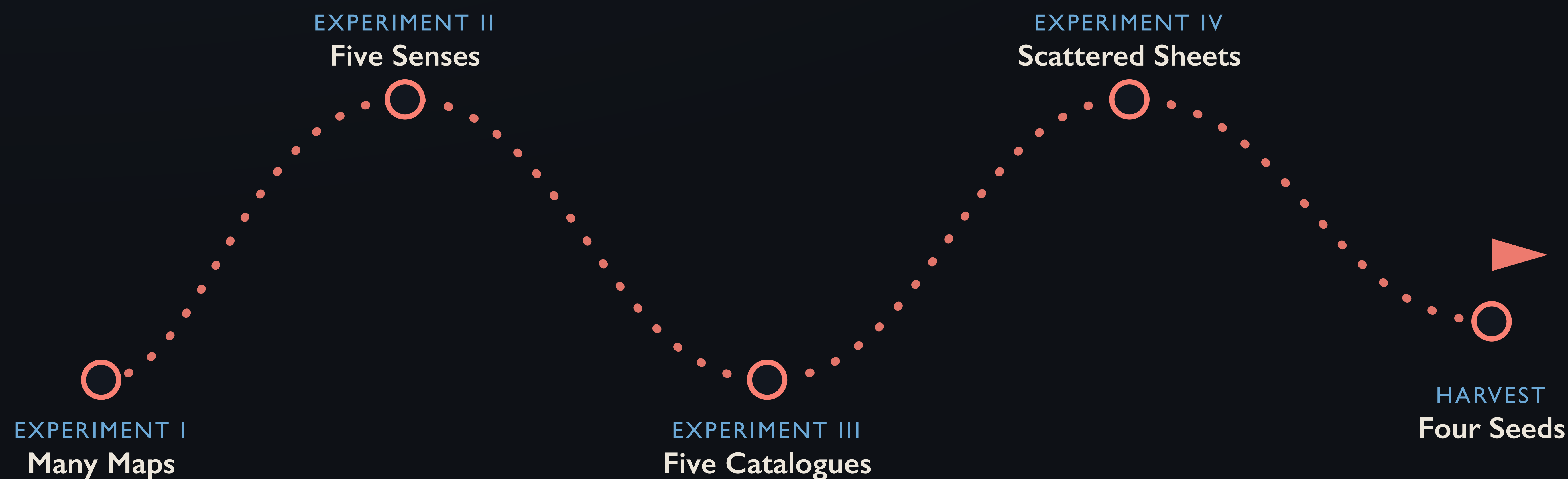
Run it, then name it

Each experiment is **run first** — with bodies, objects, and paper, no theory. Then we harvest what actually happened. Then we give the seed a quiet name and plant it for the afternoon.

*Nothing this morning requires
knowing what a retriever is.
Everything
this morning will be waiting for
you when the main deck names it.*

THE MAP FOR THIS PRELUDE

Four experiments



Many maps of one country

Four maps of the same region are on the front table: a **nautical chart**, a **road atlas**, a **subway map**, a **topographic sheet**. Each of you takes one — you may not trade.

Question one: How do I get from the station to the harbour?

Answer from your map only. Time yourself.

Question two: Where will the water go when it floods?

Whose map won — and when?

The subway map answered question one in seconds. The nautical chart was useless for it. The road atlas was right, but slow.

Then question two — and the topographic sheet, mute a moment ago, was the only map in the room that could speak.

*Nobody held a bad map. Each held a map that had **decided what to attend to** — and the deciding is what made it usable. Carroll's mile-to-the-mile map answers everything and can never be unfolded.*

A representation is a choice of what to ignore

Every map is a **lossy projection** of the country. Its power comes from the loss. The subway map threw away geography and kept topology; the topographic sheet threw away roads and kept height.

*This afternoon: BM25 keeps the words; the dense lane keeps the meaning; the factoid index keeps the claim; the summary keeps the theme. **None is the country.** We will build the atlas — and a court to decide which page to open.*

Five senses, one object

Five volunteers. One covered object.

Each volunteer gets **one channel**

only: touch for ten seconds · listen
while it is tapped and shaken ·

smell · ask three yes/no questions ·

see for two seconds.

*Each reports to the room. **Report only
what your channel gave you.***

*Then the room, together, writes one
description on the board.*

Five confident shadows

Every report was partial and every report was **certain**. The toucher knew weight and texture; the listener knew hollowness; the two-second seer knew colour and shape — and missed the weight entirely.

The board's description was better than any single report.

*Notice what did **not** happen: the seer was not the "best sensor" who made the others redundant. Each channel **lost different things** — and that is precisely why the fusion recovered more than any part.*

The tomographic principle

A CT scanner never photographs the tumour.
Hundreds of lossy shadows,
each blind alone, **reconstruct** what no single
exposure contains.

You just were the scanner. Five projections; the
whiteboard was the
reconstruction.

*This afternoon: each index is a
projection of the corpus at its own
angle, and **fusion is the
reconstruction**. The kin: the
compound
eye, the cubist portrait,
Rashomon. What it is not: the
panopticon
— one eye claiming total sight.*

EXPERIMENT II · A QUIET COROLLARY

The dragonfly's eye

Thirty thousand ommatidia, each nearly blind — and the dragonfly out-detects any human at the one thing that matters to it: **motion**. Ninety-five percent of its hunts succeed.

Why does the fusion beat the parts? Because the facets **fail independently** — each blind spot is covered by a neighbour that is not blind there.

*Hold this for the afternoon: many projections beat one **only when their failures are uncorrelated**. Five listeners would not have improved on one. Five **different** senses did. That is the wisdom of crowds — and its fine print.*

Five catalogues of one document

Every table gets the **same one-page document**. Table 1 indexes it by **author and title**. Table 2 by **subject headings**. Table 3 by **every number in it**. Table 4 by **every question it answers**. Table 5 by **one sentence — what it is about**.

*Now five queries, one at a time. For each:
which table can answer it
from its index alone — **without re-reading the document?***

EXPERIMENT III · THE FIVE QUERIES

Point, contextual, thematic, global, verbatim

1. Who wrote it?
2. What figure does it give for X?
3. What is it about?
4. Does it answer whether Y is true?
5. Quote its exact sentence on Z.

*Every query delighted **one** table
and embarrassed **four**. No table
answered all five. And query five
— the verbatim one —
embarrassed
everyone but the table that had
kept the words themselves.*

The portfolio, and the altitude of a question

Questions arrive at different **altitudes** — a name, a fact, a theme, a truth-check, a quotation — and each altitude wants a different projection. Otlet's clerks kept five repertories for exactly this reason.

*This afternoon: the **portfolio of representations** — sparse, dense, factoids, QA pairs, summaries — and the router that sends each question to its own height. You just were the Mundaneum: five drawers, one corpus, and no ranking.*

Twenty sheets, scattered

Around the room are **twenty sheets of paper** — under chairs, on windowsills, taped beneath tables. Twelve carry pieces of one argument. Eight are **decoys**: beautiful, plausible, and beside the point.

Ninety seconds. Teams of four. Collect as many sheets as you can. Do not read them yet. Go.

The synthesizing question

Back at your tables. Now read. The question every team must answer on one page:

Show that softmax, the Boltzmann–Gibbs distribution, cross-entropy loss, and logistic regression are one thing wearing four costumes — and name the thing.

Fifteen minutes. Use only the sheets on your table.

Who missed a sheet?

Some team came back without the **log-partition** sheet, or without the **sufficient-statistic** sheet — and however hard they thought at the table, the argument had a hole exactly that shape.

No amount of cleverness at the table recovered what the scoop had missed.

*That is the **recall ceiling**. What the first stage does not catch, no later stage can restore. The bucket decides what is in the sieve.*

Recall first — get it right — then spend on precision.

Who wasted time on a decoy?

Some team spent five of their fifteen minutes on the Gaussian written in full, or the lovely lemma from measure theory — decoys that sounded right and cost table time.

But notice: **the decoys cost you at the table, not at the scoop.**

Grabbing them cost nothing. Reading them cost minutes.

*That is why the pipeline is a **cascade**: scoop wide and cheap and tolerate sand; then sieve at leisure with an expensive judge. Anyone who tried to be precise while scooping came back with fewer sheets.*

Scoop wide, sieve hard

Which sheets did **every** team need? Those are your critical-path artifacts. Which sheets did every team **find** easily? Those are your cheap lanes. Which were hidden under chairs? Those are the ones a good portfolio manufactures on purpose.

*This afternoon: the **judicial ladder** — cheap judges narrow the docket so the expensive judge can afford to be careful; the archaeology of top- K that asks which artifacts actually won; and the exponential family, which you have now assembled by hand.*

What you did before lunch

- **Many maps** — a representation is a choice of what to ignore
- **Five senses** — projections fuse into more than their parts, when their blind spots differ
- **Five catalogues** — one corpus, many drawers; each question has an altitude
- **Scattered sheets** — recall first; precision is spent afterward

Four seeds. This afternoon the main deck names them: the tomographic principle, the portfolio, the recall ceiling, and the court. You will recognize each one — you were standing in it.

Now — the library

Otlet built this architecture by hand in 1895,
with clerks and index
cards, and had every drawer but the court.
This afternoon we mechanize
his library, and we build the half he never
had.